One goal, two ways to find relations

Both methods collect equations, then recover the discrete log with sparse linear algebra.

THE TARGET

$Q = kP$

Unknown: k modulo the subgroup order n

P and Q are points or divisor classes, depending on the group.

	1 CHOOSE A FACTOR BASE	2 COLLECT RELATIONS	3 SAVE EQUATIONS
ELLIPTIC $E(\mathbb{F}q)$ group elements	Selected points F_j with restricted coordinates	Solve for a point decomposition using summation polynomials	$R = \Sigma\, e_j F_j$ Check lifts and signs
HYPERELLIPTIC $Jac(C)(\mathbb{F}q)$ group elements	Selected prime-divisor classes often from rational points	Factor a divisor’s polynomial and test factor-base membership	$D = \Sigma\, e_j F_j$ Merge partial rows

Write $R = aP + bQ$.

ℓ_j: log of F_j to base P.

SAME LINEAR-ALGEBRA STAGE

$a_i + b_i k = \Sigma\, e_{ij} \ell_j \pmod n$ → **recover k**

The factor base is chosen in, or projected to, the target prime-order component. Σ means a group sum.

Testing smoothness and solving decompositions

The relation-finding step is where the elliptic and hyperelliptic methods differ most.

HYPERELLIPTIC

Test a candidate divisor

D is represented by (u(X), v(X))

Typically deg u ≤ g; g is the genus.

$$u(X) = (X - r_1)(X - r_2)$$

Example: a divisor with two rational constituents

D_1
$(r_1, v(r_1))$

D_2
$(r_2, v(r_2))$

$$[D] = D_1 + D_2$$

SMOOTH WHEN

Every constituent class belongs to the chosen factor base.

ELLIPTIC

Search for a decomposition

Seek $R = \varepsilon_1 P_1 + \cdots + \varepsilon_m P_m$

Each $x(P_i)$ must lie in a selected set V.

x_1 ? x_2 ? … x_m ?

$$S_{m+1}(x_1, \ldots, x_m, x(R)) = 0$$

Solve the polynomial system with $x_i \in V$.

Lift x-coordinates to points,
then verify the signed group sum.

Writing S is easier than solving its constrained roots.

Fast smoothness testing still needs enough relations over a sufficiently small factor base.

Divisor classes use an infinity/base-divisor convention. The displayed split assumes valid points and factor-base membership.

A cover changes the ambient group

The curve map and its induced pullback point in opposite directions.

CURVES

Genus-2 hyperelliptic curve C

$Y^2 = X^6 + aX^2 + b$

Same field $\mathbb{F}p$ as the challenge curve

degree-2 cover f

$(X,Y) \mapsto (X^2,Y)$

Elliptic curve E

$y^2 = x^3 + ax + b$

The published ECCp-131 curve

GROUPS OF DIVISOR CLASSES

Jac(C)(𝔽p)

Ambient group size ≈ p^2

Embedded elliptic subgroup

$f^*(Q) = k\, f^*(P)$

Subgroup order remains n ≈ p

pullback f*

E(𝔽p)

$Q = kP$

Challenge subgroup order n

For ECCp-131: #E(𝔽p) = n

Pushforward after pullback gives multiplication by 2.

Since n is odd, pullback is injective on the target subgroup. The same unknown k survives.

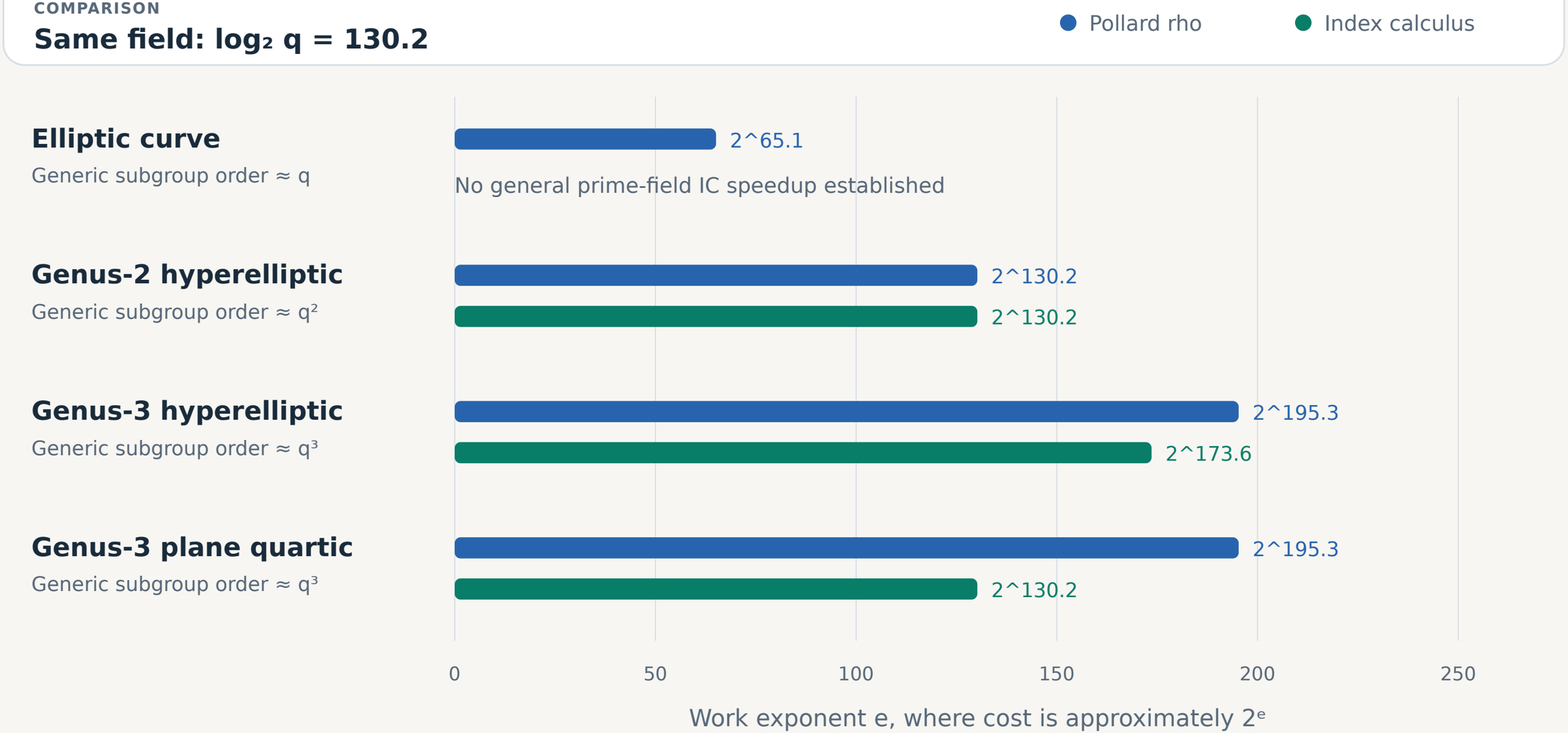
Schematic group regions, not a plot of finite-field points. A larger ambient group alone supplies no speedup.

Compare field size and group size separately

Asymptotic work exponents for representative algorithms; constants and logarithmic factors are omitted.

COMPARISON

Same field: $\log_2 q = 130.2$



These rows assume large generic subgroups. A cover of ECCp-131 retains its original order-n subgroup.

Fixed small genus and standard algorithmic hypotheses. Field/group operations and constants differ; this is not a timing prediction.